Mathematics - Graduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Ramanujan and Hardy played a game where they both picked a complex number. If the product of their numbers was $32 - 8i$, and Hardy picked $5 + 3i$, what number did Ramanujan pick?

   (a) 8-2i
   (b) 4-i
   (c) 5+3i
   (d) 1+1i
   (e) 7-5i

2. Coloring the edges of a complete graph with n vertices in 2 colors (red and blue), what is the smallest n that guarantees there is either a 4-clique in red or a 4-clique in blue?

   (a) 12
   (b) 15
   (c) 14
   (d) 16
   (e) 22

3. RS is the midsegment of trapezoid MNOP. If MN = 10x+3, RS=9x-1, and PO = 4x+7, what is the length of RS?

   (a) 20
   (b) 30
   (c) 31
   (d) 34
   (e) 32

4. In how many ways can a group of 9 people be divided into 3 non-empty subsets?

   (a) 2500
   (b) 560
   (c) 1512
   (d) 4096
   (e) 3025

5. Find the smallest positive integer that leaves a remainder of 5 when divided by 8, a remainder of 1 when divided by 3, and a remainder of 7 when divided by 11.

(a) 103
(b) 100
(c) 123
(d) 300
(e) 315

True / False (10 marks)

Answer True or False

6. True or False: The bakers can make a total of 37 apple pies using 270 apples from 15 boxes, each containing 18 apples.

7. True or False: The slope of the line passing through the points (5, 4) and (-2, 3) is 0.5.

8. True or False: The value of y that satisfies the equation y + 2.9 = 11 is 11.1.

9. True or False: In a complete graph with 22 vertices, it is guaranteed to find either a 4-clique in red or a 5-clique in blue.

10. True or False: The value of the integral $\int_C \frac{dz}{z(z-2)^2}$ where $C : |z-2| = 1$ is -0.3926.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. If $x = 2$ and $y = 5$, then what is the value of $\frac{x^4+2y^2}{6}$?

12. Given that the group (G, 0) is abelian, analyze and prove the equation (g o h)ˆ2 = gˆ2 o hˆ2 for all elements g, h in G. Discuss its implications for the structure of G.

End of Paper